

Enhanced Private Pilot Certification Course**LESSON 109 (0.7 GROUND and 1.3 DUAL)****Lesson Objective**

During this lesson, the student will become familiar with the fundamentals of flight in the airplane. This will include demonstrating various turns, climbs and descents with various power settings with little instructor guidance. This will solidify the students' understanding of how to enter and level off from climbs and descents as well as how to roll out from turns. The student will also review previously learned maneuvers for proficiency.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
 - Ask the student if they have any questions on climbing, turning, and descending procedures.
- Student questions.

References

- Airplane Flying Handbook. ([FAA-H-8083-3C](#))
 - Chapter 03 - Basic Flight Maneuvers.
- Pilot's Handbook of Aeronautical Knowledge. ([FAA-H-8083-25B](#))
 - Chapter 07 - Aircraft Systems.
- Warrior III Pilot's Information Manual.
 - Section 07 - Description and operation of the airplane and its systems.
- Warrior Maneuvers Manual.

Lesson Content (Ground)**Review Fundamentals from Lesson 108**

- Quiz student on climbing, descending, and turning procedures from previous AATD lesson.

Climbing

- Review how to enter a climb, maintain a climb, and level off from a climb.
- Climbs:
 - Entry:
 - Power.
 - Pitch.
 - Rudder.
 - Trim.
 - Maintaining:
 - Lookout.
 - Attitude indicator.
 - Airspeed indicator.

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- Altimeter.
- HSI.
- Maintain coordination as needed.
- Leveling off:
 - Anticipate by 10% of VSI.
 - Pitch.
 - Power.
 - Trim.

Descending

- Descents:
 - Entry:
 - Power.
 - Pitch.
 - Rudder.
 - Trim.
 - Maintaining:
 - Lookout.
 - Attitude indicator.
 - Airspeed indicator.
 - Altimeter.
 - HSI.
 - Maintain coordination as needed.
 - Leveling off:
 - Anticipate by 10% of VSI.
 - Pitch.
 - Power.
 - Trim.

Turning

- Medium Level Turns:
 - Entry:
 - Lookout.
 - Bank - ailerons.
 - Coordination - rudder.
 - Back pressure - stabilator.
 - Maintaining:
 - Lookout.
 - Attitude indicator.
 - Airspeed indicator.
 - Altimeter.
 - Coordination.

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- Leveling off:
 - Anticipate by $\frac{1}{2}$ bank angle.
 - Roll the wings level.

Operation of Systems - Pitot Static

- Provides pitot and static pressure to:
 - Airspeed indicator (pitot and static).
 - Altimeter (static).
 - Vertical speed indicator (static).
- Pitot mast:
 - Carries pitot pressure through lines in the wings and the fuselage.
 - A heated pitot head alleviates problems with icing and heavy rain.
- Pitot air:
 - Ram and Static air:
 - Ram air is the air coming into the pitot tube.
 - Static pressure is the ambient pressure of the outside air.
 - Airspeed indicator:
 - Diaphragm.
 - Altimeter
 - Aneroid wafers.
 - Vertical speed indicator:
 - Diaphragm.
 - Calibrated leak.
- Static Port:
 - Carries static pressure through fuselage to instruments.
- Drain Valves:
 - Drain moisture from both pitot and static lines.
 - Part of the preflight inspection!
- Alternate static source
 - Used in case the static source gets clogged.
 - Control valve installed below the left side of the instrument panel.
 - When twisted, the 3 instruments use cabin air for static pressure.
 - Storm windows and vents must be closed.
 - Cabin heater and defroster must be on.
- Blockages:

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Situation / Instrument	Airspeed	Altimeter	VSI
Blocked pitot	Zero	Works	Works
Blocked pitot and drain hole	High in climb Low in descent	Works	Works
Blocked static	Low in climb High in descent	Frozen	Frozen
Alternate static	Reads high	Reads high	Momentary climb

Go/no-go Decision

- The student must make a go/no-go decision depending on:
 - Personal minimums.
 - FIT minimums.
 - Aircraft.
 - Airspace.
- Review the students planning with a focus on:
 - Weather, weight and balance, performance, flight planning, and fuel calculations.
 - NOTAMs, TFRs, FIFs, aircraft documents, and personal documents.
 - Equipment requirements (e.g. flashlight, fuel card, drinking water, view-limiting devices, life jackets, fly away kit, survival kit, etc.)
 - FRAT/TEM/safety considerations for the flight.
- While this is represented by one line item prior to departure, the go/ no-go decision is continuous and should be a consideration even once airborne.

Lesson Content (Dual)

Preflight Inspection

- Ensure that the student conducted a thorough preflight inspection.

Visual Scanning and Collision Avoidance

- Emphasize that it is the pilot in command's responsibility to avoid traffic.
- IF YOU SEE SOMETHING, SAY SOMETHING!

Review Previous Exercises as Required

- Use the appropriate lesson plan depending on what needs to be reviewed.

Entering the Climb

- Full power.
- Right rudder.
- Set pitch attitude for the assigned airspeed (use visual references and attitude indicator).
 - V_X climb: about 12.5 degrees
 - V_Y Climb: about 10 degrees.

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- Cruise climb: about 7.5 degrees.
 - Subject to change based on current conditions.

Maintaining the Climb and Corrections

- Keep the desired pitch attitude using TRIM!
- Maintain rudder input as needed.
- Maintain heading using HSI and visual reference.
- Emphasize to level off every 500 feet to scan for traffic.

Leveling Off from a Climb

- Anticipate by using 10 percent of the vertical speed indicator.
- Lower the pitch to straight and level pitch attitude.
- Reduce the power to straight and level power setting (2300 RPM).
- TRIM!

Entering a Glide Descent

- Set power to idle.
- Left rudder.
- Set pitch attitude for a glide descent.
 - Do not let the student chase the airspeed.

Maintaining a Glide Descent and Corrections

- Keep the desired pitch attitude using TRIM!
- Maintain coordination via rudder input as needed.
- Maintain heading using visual reference and the HSI.

Leveling Off from a Glide Descent

- Anticipate by using 10 percent of the vertical speed indicator.
- Increase the pitch to straight and level pitch attitude.
- Increase the power to straight and level power setting (2300 RPM).
- TRIM!

Entering Medium Level Turns

- Check for traffic!
- Bank using the ailerons and hold a 30-degree bank (show where that is on the attitude indicator).
- Maintain coordination using the rudder.
- Increase back pressure.
 - More bank → more horizontal component of lift → less vertical component of lift → more back pressure required.

Maintaining Medium Level Turns and Corrections

- Maintaining the turn:
 - Lookout.
 - Refer to attitude indicator, airspeed indicator and altimeter.
 - TRIM for the increased back pressure.
- Corrections:

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- Change bank angle as needed.
- Hold the correction.

Rolling Out from Medium Level Turns

- Anticipate by $\frac{1}{2}$ of the bank angle.
- Roll the wings level - smoothly.
- Decrease pitch attitude.

Postflight Procedures

- Ensure that the student conducts a thorough postflight inspection.

Post-Brief

- Student self-critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 110.
- Assign homework:
 - Warrior Maneuvers Manual: normal/crosswind takeoff, power-off stalls, traffic pattern (towered airport), go-around, and normal/crosswind landing.
 - Read chapter 8 of the Airplane Flying Handbook - Airport Traffic Patterns.

Completion Standards

This lesson is complete when the student is able to:

1. Correctly perform previously learned maneuvers with little instructor guidance.
2. Demonstrate proper sequential checklist procedures for before and after start, before and after taxi, run-up, takeoff, climb and cruise without instructor guidance.
3. Demonstrate how to enter and level off from climbs and descents as well as how to enter and roll out from turns with little instructor guidance.
4. Control rate of descent when performing a glide descent by utilizing the manufacturers recommended glide speed with little instructor guidance.
5. Explain the effect of altitude and wind on glide distance.